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Description - Contains theory for the SPI and SPU portions of IPG and IPG II Boards.

1- IPG AND IPG II SERIAL PERIPHERAL INTERFACE

The Serial Peripheral Interface (SPI) section of the IPG Board is responsible for handling serial I/O to peripherals described below:

- Operator Work Station
- SRI (Scan Room Interface)
- MDS (Multidrop Serial Link) which communicates with:
 - Gradient Amplifier Serial to Parallel Converter
 - T/R Driver & RF Amplifiers Serial to Parallel Converter
 - Signal Collecting Board
- Debug Terminal

The SPI consists of an Intel 80C196 8/16 bit microcontroller running at a clock speed of 11.1MHz, along with the following peripherals:

- One 1Kx16 DPR - Dual Port RAM (SPI - 68020)
- One 1K x 8 DPR - Dual Port RAM (SPI - SPU)
- Two DUARTs
- Two 32K * 8 CMOS Static RAMs - 32K x 16 configuration
- Computer Operating Properly Indicator - green led
- Debug Terminal Port

1-1 68020 Interface

The SPI interfaces to the 68020 CPU through a 2Kx16 Dual Port RAM (the RAM size is only 1Kx16 in RUN mode). Both the 80C196 microprocessor used in the SPI and the 68020 have access to the RAM. The SPI interrupts the 68020 by pulsing the P2.5 pin on one of the 68020 MFP chips; the SPI interrupts on the P0.7 pin.

1-2 SPU (Signal processing Unit) Interface

In addition to the 68020 interface, the SPI also has a 1Kx8 DPR interface to the SPU CPU. Hardware guarantees that no bus contention occurs and the RAM generates interrupts for each processor.

1-3 SPI Main Memory

In addition to the dual port RAMs, the SPI also has 32K x 16 of SRAM. Since the SPI's CPU can only access 64K bytes of RAM, decoding is set up so part of the 32K x 16 RAM is mapped out to make room for the SPI's peripherals.

1-4 MDS (Multidrop Serial) Interface

The MDS communication link is handled by the internal serial port on the SPI CPU.

1-5 SRI (Scan Room) Interface

The Scan Room Interface is handled by the A channel of DUART1. The serial channel operates at 19.2K baud.

1-6 Debug Interface

The debug interface is handled by channel A of DUART2 . This serial channel operates at 9600 baud. Channel B of DUART2 is not used.

1-7 SPI Interrupt Controller

An interrupt controller (contained in an EPM5128 MAXPAL) is provided for the 80C196. The interrupt controller is connected to the EXTINT interrupt pin. The other interrupt pin, P0.7, is connected to the 68020-SPI Dual Port Ram. The interrupt controller has 8 inputs as shown below in order of priority (8 = highest):

8 (SPU DPR) - SPU-SPI 1K x 8 Dual Port Ram Interrupt

7 (CPB TXD) - Transmit data interrupt for CPB

6 (Debug RXD) - Receive data interrupt for Debug Terminal

5 (SRI RXD) - Receive data interrupt for SRI

4 (CPB) - Receive data interrupt for CPB

3 (LCA SSI) - SSI Interrupt for Pulse Controller

2 (Debug TXD) - Transmit data interrupt for Debug Terminal

1 (SRI TXD) - Transmit data interrupt for SRI

1-8 Test Register

The Test Register provides the following functions:

Bit 3: Serial Channel Loop Back Control on IPG

Bit 0: Scan Room Interface (SRI) Reset

Bit 2: MDS Peripheral Reset - This resets all peripherals on the link.

Bit 1: SPI Operating Properly LED Indicator - This bit is toggled on and off once every 200ms.

Bit 7: Run Mode - When this bit is set to one, the SPI-68020 DPR is remapped into the RUN position. Once this bit is set, it can only be reset by a processor reset.

2- IPG SIGNAL PROCESSING UNIT (SPU)

The SPU is used to process physiological signals sent to it from the PAC (physiological acquisition controller) by a high-speed serial link. It is also used to process waveforms sent to it by the 68020. The SPU uses a TMS320C30 digital signal processor running at a clock speed of 33.33MHz. The major features of the SPU are:

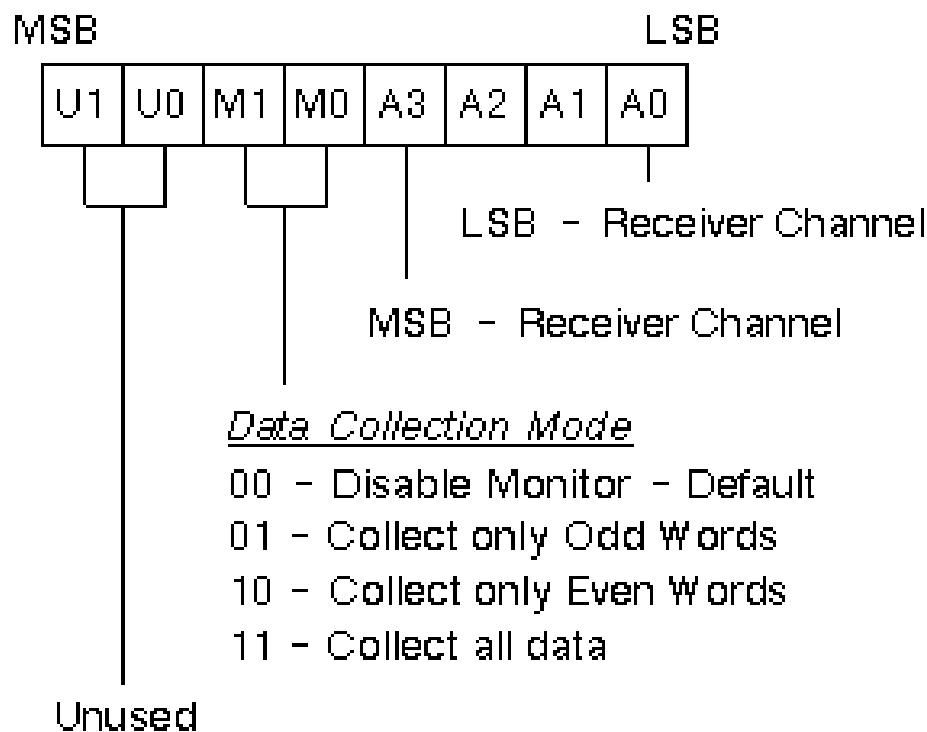
- TMS320C30 DSP
- 2K x 32 1/3ws IDT7132SA Dual Port SRAM on Main Bus - 68020-SPU
- 64K x 32 0 Wait State dedicated SRAM
- High Speed Serial Channel to TYME Board or Output of Physiological Waveforms
- High Speed Serial Channel (345K Baud) to PAC
- 1K x 8 1/3ws IDT7130SA Dual Port RAM on Extended C30 Bus - SPI-SPU
- Pulse Controller Interface
- Multiple Interrupts
- XD Bus Monitor with 256 x 16 FIFO with half full indicators

2-1 68020 Interface

The SPU does not have any boot PROM. Instead, the SPU is held reset by the 68020 at power up. The 68020 loads the 2K x 32 DPR, which is on the DSP's main bus, with boot code and releases the reset line on the SPU. The DPR hardware guarantees that no bus contention occurs even with simultaneous accesses to the same address.

2-2 XD Bus Monitor

The XD Bus Monitor captures information being sent from any one of the 16 receiver channels (with the maximum four Receiver Boards, four channels each). An address compare register is provided to hold the address of the receiver channel that is to be monitored. In addition, the 4th and 5th bits of the address compare register are used to specify which data to capture. See Illustration L1955A.



L1955A

ADDRESS COMPARE REGISTER
ILLUSTRATION L1955A

The XD monitor latches the appropriate data into a 256 x 16bit FIFO and interrupts the SPU CPU on interrupt line INT0*. The interrupt sent to the C30 is a logical-OR of FIFO full, FIFO half full, and FIFO empty. The bits 0 and 1 of the interrupt status register are used to differentiate between interrupts. These interrupts are latched when they occur and are not be cleared until the XD Bus Status Register is read. Shown below is the truth table for the XD Bus Status Register:

<u>XS1</u>	<u>XS0</u>	<u>Interrupt Source</u>
0	0	FIFO Empty
0	1	FIFO Half Full
1	0	FIFO Full
1	1	Not Used

2-3 SPU CPU I/O Flags

The XF1 flag is used as an input for the SPU CPU lock instruction to synchronize accesses to the WARP X, Y, and Z registers. The XF1 pin status can be read from bit 7 of the IOF register. At reset, the C30 sets the XF pins to act as inputs, so no additional setup is required.

2-4 PAC Serial Port

This serial port sends and receives data from the PAC. Since the serial ports on the SPU CPU do not support asynchronous communications, additional hardware is used to implement the asynchronous format. The format for the serial link to the PAC is 346.8K baud with 1 start bit, 8 data bits, and 2 stop bits. The internal serial port on the SPI CPU is used to transmit and receive data, however, external hardware does the conversion from synchronous to asynchronous.

2-5 PAC Serial Port Loop Back

The PAC Serial Port can be looped back on the IPG Board by setting bit 3 in the SPI Test Register.

2-6 TYME II Board Serial Port

The serial port to the TYME II board is 32-bit serial synchronous format. The SPU CPU sends data to the TYME II board at a rate of about 32 Kbits per second. At this data rate, 32 bits are written every millisecond, or at a rate of 1 kHz. This channel is used to send digital physiological waveforms to the TYME II board, where DACs are used to produce analog waveforms. The 32-bit word is broken up into four 8-bit bytes that go to four 8-bit DACs.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.